Sign Language to Smart-Glasses HUD: Real-Time ISL Word Display on an ESP32-C3 OLED Head-Up Display

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Declaration

In this submission, we have expressed our ideas in our own words, and we have adequately cited and referenced any ideas or words that were taken from another source. We also declare that we adhere to all principles of academic honesty and integrity and that we have not misrepresented or falsified any ideas, data, facts, or sources in this submission. If any violation of the above is made, we understand that the institute may take disciplinary action. Such a violation may also engender disciplinary action from the sources which were not properly cited or permission not taken when needed.

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To Whom It May Concern

This is to certify that the project report entitled **"Sign Language to Smart-Glasses HUD: Real-Time ISL Word Display on an ESP32-C3 OLED Head-Up Display"** submitted to the Department of Electronics and Communication Engineering, Indian Institute of Information Technology Senapati, Manipur in partial fulfilment for the award of degree of **Bachelor of Technology in Electronics and Communication Engineering** is a record of bonafide work carried out by **Ayush Verma** bearing roll number **230102017** and **Charan Tej** bearing roll number **230104002** for the course **Project II (EC4201)**.

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ABSTRACT

People who use Indian Sign Language (ISL) are often not understood by people who do not sign. This project builds a low-cost wearable aid for the listener: a pair of glasses with a small head-up display (HUD) that shows, in real time, the words a signer is making. A laptop camera films the signer. A recognition application extracts hand and body landmarks with MediaPipe, classifies each sign with a temporal convolutional network (TCN) and builds a sentence. Each word is sent as a small JSON packet over UDP on a phone Wi-Fi hotspot.

On the glasses, an ESP32-C3 SuperMini microcontroller joins the hotspot, registers the mDNS name `glasses.local`, receives the packets on port 4210 and draws the text on a 0.96-inch SSD1306 OLED. A short word is shown large in the centre; a sentence scrolls across the screen twice. The OLED image passes through a 25 mm lens and is reflected into the eye by a 45° clear acrylic sheet, so the text floats in front of the wearer. The unit runs from a 600 mAh LiPo cell through a TP4056 protection board, and a parametric OpenSCAD enclosure that clips onto the glasses arm was designed and exported for 3D printing.

Every part was verified with a step-by-step bring-up procedure, and the link was tested end to end with typed text, simulated signing and live signing. The bench optics gave a sharp image at an OLED-to-lens distance of 22 mm, and the measured sentence timing (14.6 s) matched the model (14.67 s). The work combines embedded firmware, wireless networking, optics, power electronics, mechanical CAD and machine learning in one working system.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
BiLSTM	Bidirectional Long Short-Term Memory
CAD	Computer-Aided Design
CC/CV	Constant Current / Constant Voltage (charging)
DIY	Do It Yourself
ECE	Electronics and Communication Engineering
GPIO	General-Purpose Input/Output
GPU	Graphics Processing Unit
HUD	Head-Up Display
I2C	Inter-Integrated Circuit (two-wire serial bus)
IP	Internet Protocol
ISL	Indian Sign Language
JSON	JavaScript Object Notation
LED	Light-Emitting Diode
LiPo	Lithium Polymer (battery)
mDNS	Multicast Domain Name System
ML	Machine Learning
OLED	Organic Light-Emitting Diode
PLA	Polylactic Acid (3D-printing plastic)
SCL	Serial Clock Line
SDA	Serial Data Line
STL	Stereolithography (3D mesh file format)
TCN	Temporal Convolutional Network
TX	Transmit
UDP	User Datagram Protocol
USB	Universal Serial Bus

LIST OF SYMBOLS

Symbol	Description
u	Object distance, OLED screen to lens centre (22 mm)
v	Image distance from the lens (mm; negative = virtual image)
f	Focal length of the lens (estimated ≈ 25 mm)
m	Lateral magnification of the HUD image
n	Number of characters in a displayed sentence
t_{pass}	Time for one full scroll pass of a sentence (s)
t_{total}	Time a sentence stays on screen, 2 passes (s)
V_{5V}	Voltage at the ESP32-C3 5V pin (V)
V_{3V3}	Voltage at the ESP32-C3 3V3 pin (V)

CHAPTER 1

INTRODUCTION

1.1 Overview

- Indian Sign Language (ISL) is the main language of the Deaf community in India.
- Most hearing people do not know ISL, so everyday conversations need a human interpreter.
- Computer vision can now track hands and body in real time on an ordinary laptop camera.
- Small, cheap Wi-Fi microcontrollers and OLED screens make a home-built head-up display (HUD) possible.
- This project combines the two: a "caption" for sign language that the listener sees in their glasses (Figure 1.1).

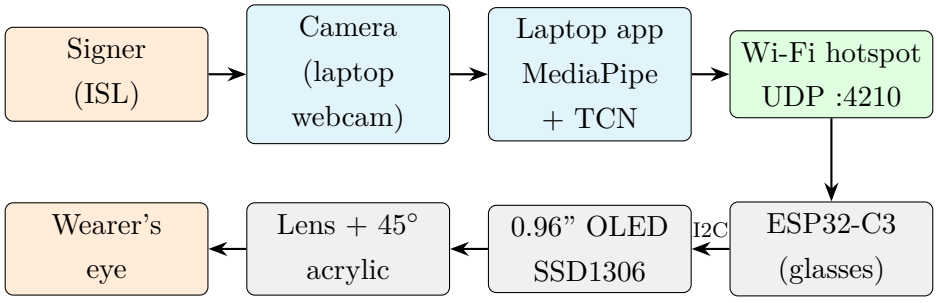


Figure 1.1: System overview: from the signer to the text in the wearer's eye

1.2 Sign Language Accessibility and Wearable HUDs

- Phone-screen captions force the listener to look away from the signer.
- Commercial smart glasses are expensive and closed.
- A low-cost DIY HUD built from hobby parts can be made and repaired by students.
- Earlier DIY HUD designs used weak electronics (no regulator, ESP8266-01, hard-coded IP) [11]; this project fixes those.

1.3 Problem Statement and Key Contributions

Problem statement:

- Show each recognised ISL word to a hearing person **in their field of view**, hands-free, in real time.
- The display unit must be **light, battery powered and wireless**, and must clip onto ordinary glasses.
- The text must be **sharp and readable** through simple optics (one lens and a beam combiner).
- The glasses and the recognizer must talk over a **simple, documented network link**.

Key contributions:

- A step-by-step, pass/fail bring-up of every hardware part (ESP32-C3, OLED, charger, battery, switch).
- Wireless display firmware: mDNS discovery, UDP packets, centred words and scrolling sentences.
- A no-mirror HUD optics path, measured on the bench (OLED $\rightarrow$ 22 mm $\rightarrow$ lens $\rightarrow$ 45° acrylic).
- A parametric, print-ready clip-on enclosure in OpenSCAD.
- A sign-recognition application (MediaPipe + TCN) that sends every word and sentence to the glasses.
- A written app-to-glasses interface, tested end to end.

Table 1.1: Work division between the team members

Member	**Responsibility**
Ayush Verma (230102017)	Sign-recognition software: dataset preparation, MediaPipe landmarks, TCN model, training, live recognition app and UDP sender [12]
Charan Tej (230104002)	Hardware and firmware: ESP32-C3 + OLED display unit, LiPo power, HUD optics, 3D enclosure, glasses-side link and bring-up tests
Both	Shared UDP interface, integration and end-to-end testing, documentation

1.4 Objectives and Scope

Objectives:

1. Bring up and verify every hardware part step by step.
2. Write firmware that receives words over Wi-Fi (UDP + mDNS) and shows them clearly on the OLED.
3. Design the HUD optics: OLED $\rightarrow$ lens $\rightarrow$ 45° acrylic $\rightarrow$ eye.
4. Design a 3D-printable clip-on enclosure for the display unit.
5. Connect the glasses to the recognizer and test the full chain end to end.

Scope:

- **In scope:** isolated ISL words (INCLUDE-50 [5] and our own recorded words) and short sentences built from them; display unit for one eye (right); laptop as the recognition computer.
- **Out of scope:** continuous sentence-level sign translation; recognition running on the glasses.
- **Stretch goal:** an ESP32-CAM on the glasses as the camera.

1.5 Report Organization

- **Chapter 2** reviews existing approaches and the theory used: microcontroller, display, network, optics and battery charging.
- **Chapter 3** describes the system design and implementation: electronics, firmware, optics, enclosure, recognizer and integration.
- **Chapter 4** presents the test results, measurements and analysis, and the challenges met with their solutions.
- **Chapter 5** concludes the report, lists the limitations and suggests future work.
- **Appendices A–D** give the wiring and parts, a firmware excerpt, the UDP interface and the bring-up sketches.

CHAPTER 2

LITERATURE REVIEW AND THEORETICAL BACKGROUND

2.1 Existing Approaches and Limitations

Table 2.1: Existing ways to show sign-language output to a listener

Method	**How it works**	**Limitation**
Human interpreter	Person translates live	Not always available, costly
Phone / laptop captions	Recognised text on a screen	Listener looks away from the signer
Commercial smart glasses	Built-in micro display	Expensive, closed platform
Sensor gloves	Flex/IMU sensors on the signer's hands	Signer must wear special hardware
This project	Camera + ML on laptop, DIY HUD glasses	Needs a laptop and a hotspot

2.2 Review of Related Work

- **MediaPipe** [4]: real-time pipelines for hand and body landmark tracking on ordinary CPUs and GPUs; turns video into landmark sequences.
- **INCLUDE dataset** [5]: a large ISL word dataset (263 words, 4,287 videos); its 50-word subset INCLUDE-50 is used for training.
- **Temporal Convolutional Networks** [6]: 1-D dilated convolutions that match or beat recurrent networks on sequence tasks; our main recognition model is a TCN.
- **June_Glasses** [11]: open DIY HUD glasses; gave the optics concept (OLED, lens, beam combiner). Its electronics had no regulator and a hard-coded IP.

- **Adafruit SSD1306 / GFX libraries** [8, 9]: standard Arduino drivers for the OLED, with text, bitmaps and size scaling.
- **OpenSCAD** [10]: script-based CAD; every dimension is a named parameter, so the enclosure can be changed and re-exported quickly.

2.3 Theoretical Background

2.3.1 ESP32-C3 Microcontroller and SSD1306 OLED

- **ESP32-C3:** 32-bit RISC-V microcontroller with Wi-Fi 802.11 b/g/n (2.4 GHz only) and native USB [1].
- **SSD1306 OLED:** 128×64 monochrome display driver, controlled over I2C (address 0x3C) [2].
- **I2C bus:** two-wire serial bus (SDA data, SCL clock) between the microcontroller and the OLED.

2.3.2 Wireless Link: UDP and mDNS

- **UDP:** connectionless transport; one small datagram per word gives low delay and no connection set-up.
- **mDNS:** lets the laptop find the glasses by name (`glasses.local`) instead of a changing IP address.

2.3.3 HUD Optics and the Thin-Lens Equation

- A convex lens placed closer to the OLED than its focal length makes a magnified *virtual* image.
- A 45° clear sheet reflects the image into the eye while the world stays visible (Figure 2.1).

Thin-lens equation (real-is-positive sign convention) [7]:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}, \qquad m = -\frac{v}{u} \tag{2.1}$$

where u is the object (OLED) distance, v the image distance and f the focal length. If $u < f$, then $v < 0$: the image is virtual, upright and magnified.

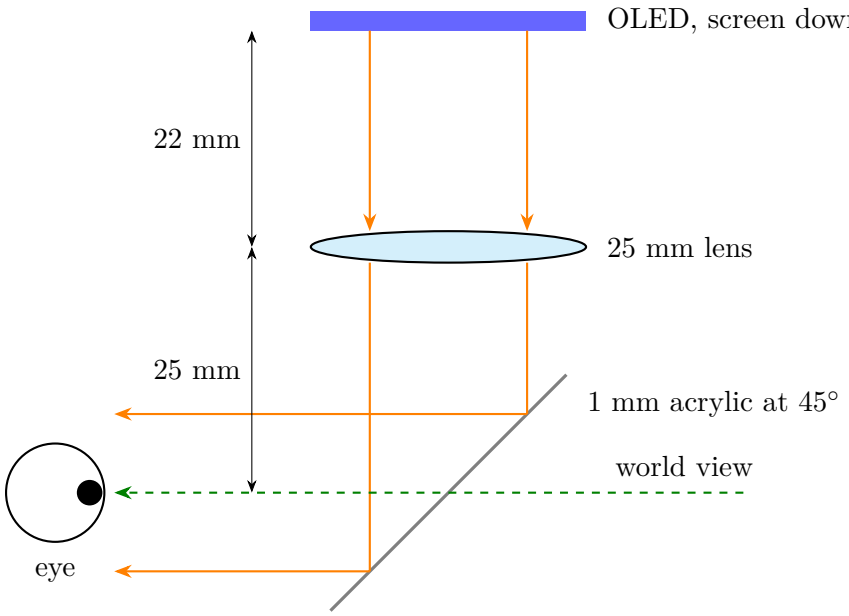


Figure 2.1: HUD light path (side view): OLED → lens → 45° acrylic → eye

2.3.4 Single-Cell LiPo Charging

- **TP4056:** single-cell Li-ion linear charger (CC/CV, 4.2 V) with a protection IC [3].
- The stock board charges at 1 A, which is 1.7C for a 600 mAh cell, so it is too high for our battery.
- Safe charging used here: a bench supply at 4.20 V with a 0.30 A limit, stopped when the current falls to $\leq$ 0.05 A.

2.4 Identified Gap and Technical Requirements

- No low-cost, open, clip-on HUD that shows **ISL** recognition output was found.
- **Requirements derived for this project:**
  - Wireless link with discovery by name (mDNS) and a fixed, small packet format.
  - Readable text: large, centred words; long sentences scroll.
  - Safe single-cell LiPo power with protection and a power switch.
  - Adjustable focus and a mount that fits a given pair of glasses.
  - Software-selectable mirroring (`MIRROR_X`, `FLIP_Y`) in case the optics flip the text.

CHAPTER 3

SYSTEM DESIGN AND IMPLEMENTATION

3.1 Design Methodology and Implementation Flow

- We followed a "test every part first" approach: each step has a clear pass/fail check.
- Hardware was tested on a breadboard with friction-fit headers; soldering is done only in the final build.
- The software and the glasses were developed in parallel against one written UDP interface.

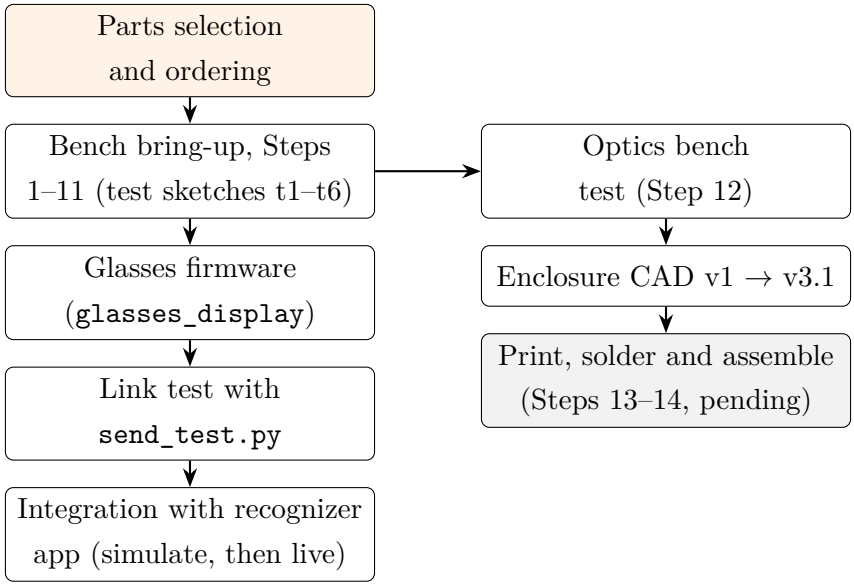


Figure 3.1: Design methodology and implementation flow

3.2 Hardware and Software Tools

Table 3.1: Hardware used for the display unit

Part	**Qty**	**Purpose / note**
ESP32-C3 SuperMini	1 (+1)	Wi-Fi microcontroller, native USB-C
0.96” SSD1306 OLED, 128×64, I2C	1 (+1)	Display, address 0x3C
LiPo 3.7 V, 600 mAh (602030)	1	Battery, ∼30×20×6 mm
TP4056 Type-C board with protection	1	Charger + over-discharge protection
Slide switch	1	Power on/off
25 mm lens (marked f45)	1	Magnifies the OLED image
Clear acrylic, 1 mm, 30×30 mm	1	45° beam combiner
PLA filament (∼73 g)	–	3D-printed enclosure
Breadboard, berg strips, jumpers	–	Bench bring-up
Bench supply, Rigol DM3058E meter	–	Charging and measurements

Table 3.2: Software and tools used

Tool	**Use**
Arduino IDE 2.3, ESP32 core 3.3.12	Firmware build and upload
Adafruit SSD1306 + GFX, WiFiUdp, ESPmDNS	OLED drawing, UDP, mDNS
Python, PyTorch, MediaPipe, OpenCV	Sign-recognition application
Python 3 + PyYAML	`send_test.py` link-test tool
OpenSCAD	Parametric 3D enclosure, STL export
Git	Version control and daily work log

3.3 Electronics and Wiring

- LiPo → TP4056 (B+/B−) → OUT+ → slide switch → ESP32-C3 `5V`; OUT− → `G`.
- ESP32-C3 → OLED: `3V3`→VDD, `G`→GND, `GPIO8`→SDA, `GPIO9`→SCK (SCL).
- The ESP32-C3's on-board regulator gives 3.3 V from the 3.7–4.2 V cell.
- **Safety:** the TP4056 USB port is taped ("NO USB"); the cell is charged only from the bench supply.

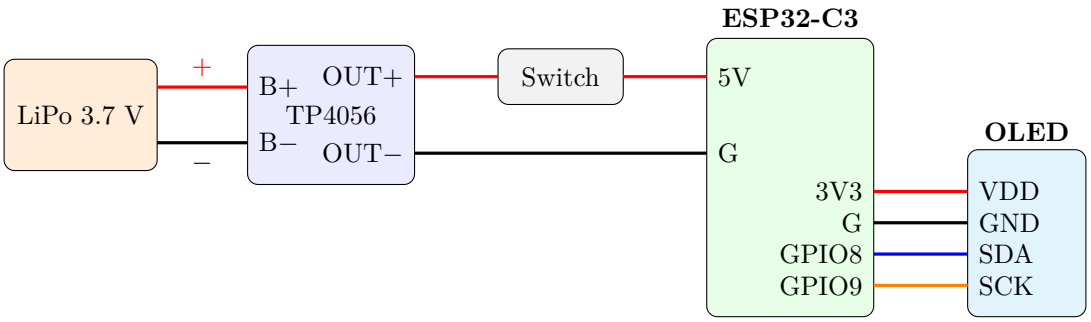


Figure 3.2: Wiring of the glasses display unit

3.4 Firmware Design

- **Boot:** start I2C and the OLED, apply `MIRROR_X`/`FLIP_Y`, show `WIFI...`.
- **Network:** join the hotspot, lower Wi-Fi TX power to 8.5 dBm (the SuperMini antenna is weak), start mDNS `glasses` and UDP port 4210, show `READY` + IP.
- **Receive:** read one datagram into a 256-byte buffer and pull the `"text"` value out of the JSON (a non-JSON packet is shown raw).
- **Display rule:** text at size 2 (12×16 px per character) that fits 128 px (≤10 characters) is centred; longer text scrolls right-to-left, 2 passes at 60 px/s, then the screen goes blank.
- Scroll position is computed from elapsed time, so the speed stays constant however long a redraw takes.

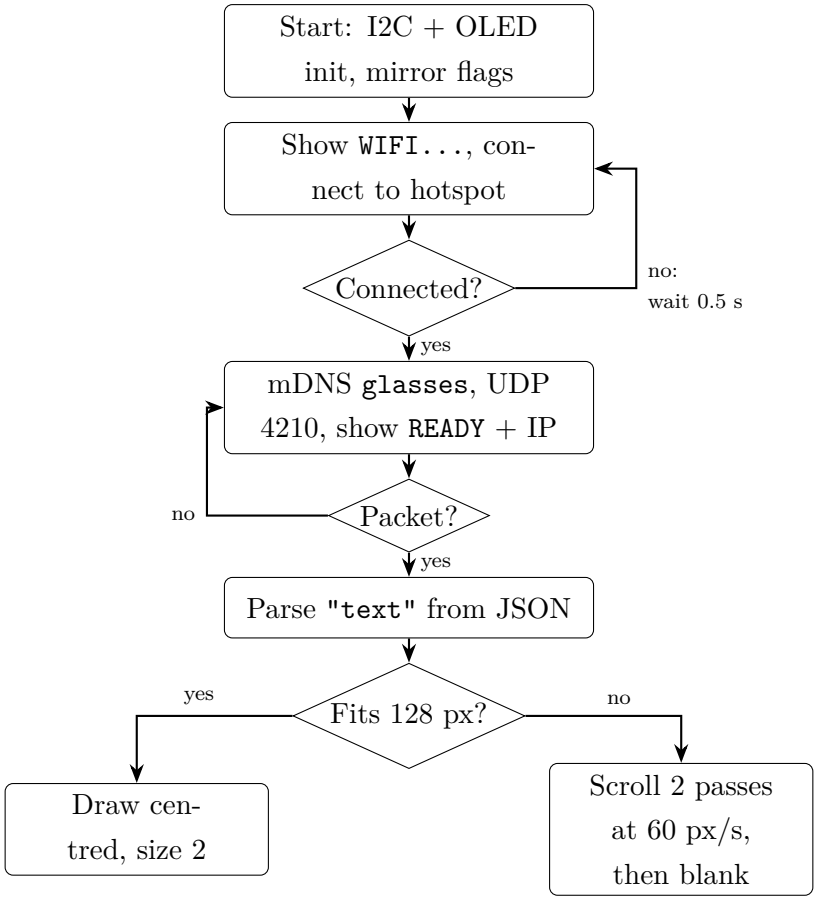


Figure 3.3: Firmware flowchart of the glasses receiver

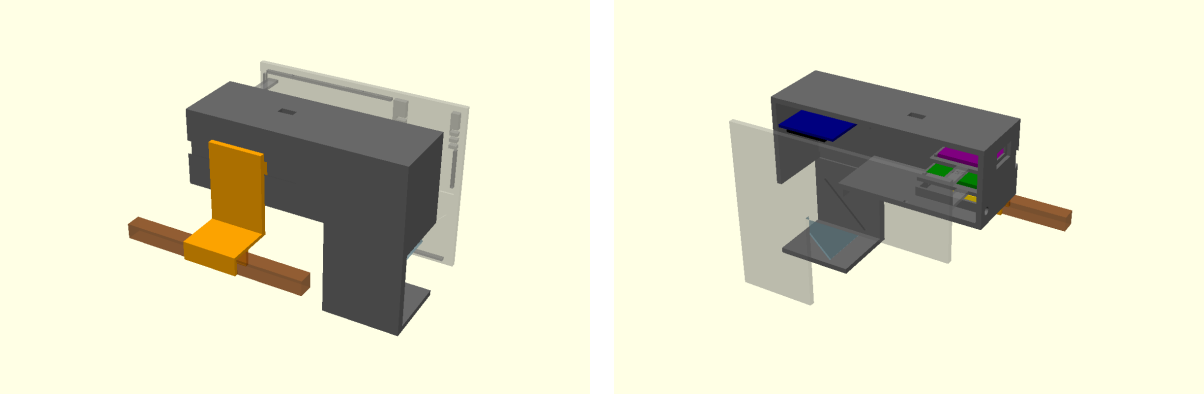
3.5 Optics Design

- The first design used a mirror and $u = 40\,\text{mm}$, based on the lens marking $f = 45\,\text{mm}$.
- The bench test (Step 12) showed a sharp image at only $u = 22\,\text{mm}$, so the mirror was removed.
- Final path: OLED (screen down) $\rightarrow 22\,\text{mm} \rightarrow$ lens $\rightarrow 25\,\text{mm} \rightarrow 45°$ acrylic $\rightarrow$ eye (Figure 2.1).
- The OLED sits in rail slots at $u = 19$, 22 or $25\,\text{mm}$, so focus can be adjusted after printing.

3.6 Enclosure Design

- Fully parametric (OpenSCAD, v3.1): optics distances, clearances, board sizes and glasses-arm sizes are named values.
- A pocket for each part (LiPo, TP4056, ESP32-C3, switch); the lid clicks shut and its fingers hold the boards.

- Clip-on mount: a dovetail slot in the body plus a swappable clip for the 7×7 mm temple arm.
- The build script stops with an error if any parts collide; the clip snap strain is checked (2.2%, below PLA's ∼2.5% limit).



(a) Assembly on the glasses arm (temple side) (b) Inside view (lid see-through)

Figure 3.4: 3D model of the HUD display unit (OpenSCAD render)

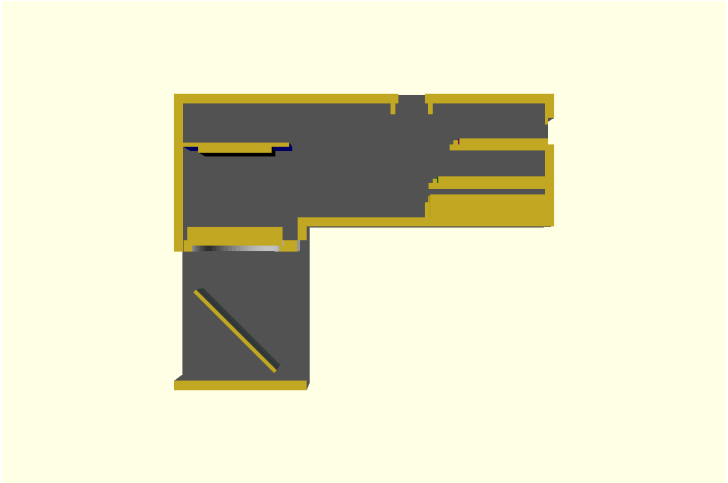


Figure 3.5: Cross-section of the enclosure: OLED rails at the top, lens plate, 45° acrylic in the head

3.7 Sign-Recognition Software

- Webcam → MediaPipe hand + body landmarks [4] → TCN classifier [6] (BiLSTM and Transformer also available) → sentence builder → optional English translation.
- Trained on INCLUDE-50 [5] and our own recorded signs; runs on the laptop GPU.
- `glasses.py` sends each word, then the finished sentence, to the glasses over UDP [12].
- A `--simulate` mode sends example sentences, so the glasses can be tested without a camera or model.

3.8 Integration and Test Procedure

1. **Step 1:** visual check; measure the board's 5V and 3V3 pins with the multimeter.
2. **Steps 2–3:** board on friction-fit headers; `t1_blink` (LED + Serial `alive N`).
3. **Step 4:** `t2_i2c_scan` (expect 0x3C), `t3_oled_hello`, mirror check.
4. **Steps 5–6:** `t4_wifi_udp`; then on the laptop:
   `python tools/send_test.py --host <IP> --text hello`.
5. **Steps 7–9:** TP4056 and switch on USB only; bench-charge the LiPo; solder the battery leads.
6. **Steps 10–11:** run the ESP32-C3 from the battery; repeat the tests on the spare boards.
7. **Step 12:** optics bench: slide the OLED until the text through the lens is sharpest; add the acrylic at 45°.
8. **Integration:** flash `glasses_display`; run `python software/run.py --simulate`, then the live camera.

Packet format (one UDP datagram per word or sentence, UTF-8 JSON, under 256 bytes):

```
{"text": "hello", "conf": 0.92}
```

3.9 Phase-wise Work Summary

Table 3.3: Phase-wise work summary

Date	Work done
24 Oct 2026	Parts received; bring-up guide and test sketches written; Step 1 passed on both boards
25 Oct 2026	Steps 3–6: blink, I2C scan (0x3C), OLED text, Wi-Fi + UDP text from the laptop
29 Oct 2026	Enclosure v1 in OpenSCAD; LiPo checked; ESP32-C3 + OLED running on the battery
30 Oct 2026	Enclosure v2: component pockets, click lid, dovetail clip mount
31 Oct 2026	Steps 7–11 done; optics bench (sharp at 22 mm); enclosure v3/v3.1 print-ready; UDP interface written
1 Nov 2026	Final receiver firmware; recognizer integrated with the glasses; simulated and live runs; documentation

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Hardware Bring-Up Results

- All bring-up tests, Steps 1–11, passed on both sets of boards (Table 4.1).
- Typed text from the laptop appears on the OLED within about 0.5 s (by eye), by IP address and by `glasses.local`.
- The whole chain works: simulated and live signing in front of the camera show words, then the sentence, on the OLED.
- Final firmware compiles to 79% of the ESP32-C3 flash.

Table 4.1: Hardware bring-up test results

Part	**Test**	**Result**
ESP32-C3 #1, #2	5V/3V3 voltages, blink + Serial	Pass
OLED #1, #2	I2C 0x3C, text, mirror view	Pass
Wi-Fi + UDP	`send_test.py` text by IP and by `glasses.local`	Pass
TP4056 #1, #2	B+ $\approx$ 4.2 V on USB (no cell)	Pass
Slide switch $\times$3	Continuity swaps when flipped	Pass
Bench supply	4.20 V / 0.30 A set and checked with the meter	Pass
LiPo #1, #2	Voltage, soldered leads, "NO USB" tape	Pass
Battery power	`READY` + words on battery, switch on/off	Pass
Battery life	Runtime on a full cell	Done (value not logged)
Optics	Sharp text through lens + acrylic	In progress
Soldering, enclosure	Final build	Pending

4.2 Electrical Measurements

Table 4.2: Measured electrical values (Rigol DM3058E)

Quantity	**Measured**
ESP32-C3 5V pin (USB), V_{5V}	5.162 V
ESP32-C3 3V3 pin (USB), V_{3V3}	3.302 V
LiPo #1 as received	3.999 V ($\sim$80%)
ESP32-C3 5V pin on battery, under load	3.967 V
Best focus, OLED to lens (u)	22 mm

- 5.162 V and 3.302 V are inside the expected 4.8–5.2 V and 3.2–3.4 V ranges.
- On battery the 5V pin reads the cell voltage (3.967 V) while 3V3 stays regulated, so the board runs safely on one LiPo cell.

4.3 Display Timing Analysis

At text size 2 each character is 12 px wide; one pass moves the text across its own width plus the 128 px screen:

$$t_{\text{pass}} = \frac{12\,n + 128}{60}\ \text{s}, \qquad t_{\text{total}} = 2\,t_{\text{pass}} \tag{4.1}$$

where n is the number of characters and 60 px/s is `SCROLL_PX_PER_S`.

- For "I am going to school." ($n = 21$): $2 \times (12 \cdot 21 + 128)/60 = 12.67$ s; plus the 2 s word gap of the simulator = 14.67 s.
- The measured gap was **14.6 s**, so the firmware and the app stay in step (Figure 4.1).

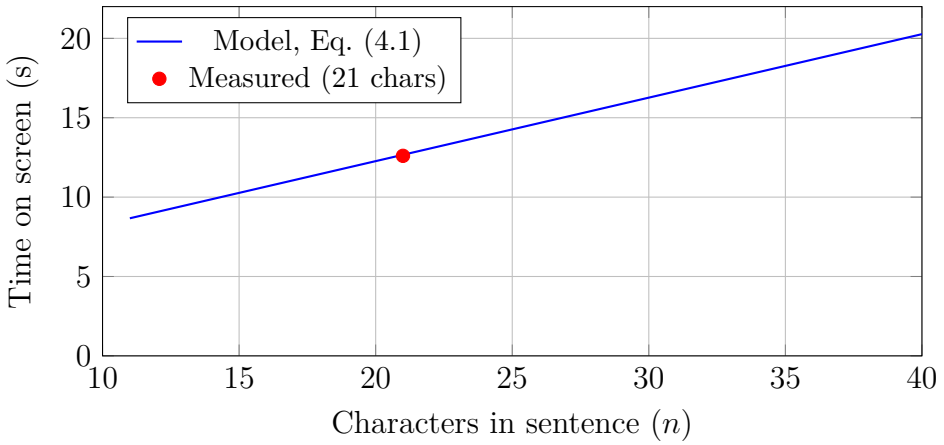


Figure 4.1: Sentence scroll time vs. sentence length (2 passes at 60 px/s)

4.4 Optics Analysis

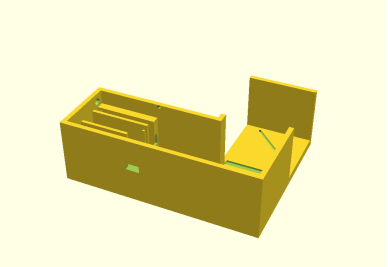
With $u = 22\,\text{mm}$ and an estimated $f \approx 25\,\text{mm}$, Eq. (2.1) gives

$$\frac{1}{v} = \frac{1}{25} - \frac{1}{22} = \frac{22 - 25}{550} = -\frac{3}{550} \;\Rightarrow\; v \approx -183\ \text{mm}, \quad m = -\frac{v}{u} = \frac{183}{22} \approx 8.3$$

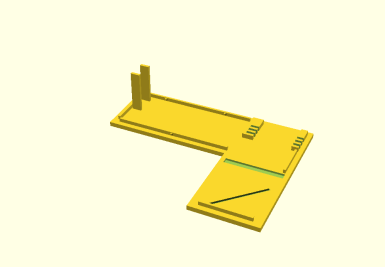
- The text appears as an upright, about 8× magnified virtual image roughly 18 cm in front of the lens.
- The value of f is an estimate; measuring it directly (Step 12, O5) is still pending.

4.5 Enclosure Results

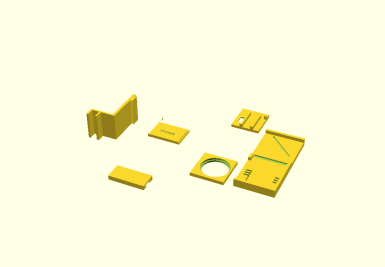
- Enclosure v3.1 is print-ready: all parts render with no collisions (Figures 4.2 and 4.3).
- All four parts fit on one print bed: about 73 g of PLA (Table 4.3).



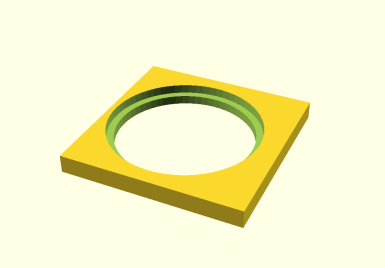
(a) Body



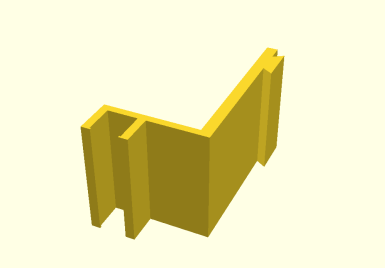
(b) Lid



(c) Test-fit piece



(d) Lens plate



(e) Clip

Figure 4.2: Printable enclosure parts (OpenSCAD renders)

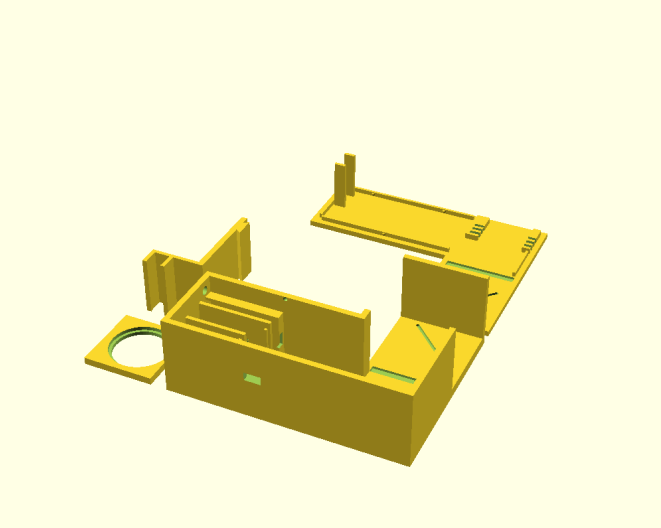


Figure 4.3: All four parts laid out on one print bed (137 × 161 mm)

Table 4.3: Enclosure parts: plastic and print time (slicer estimates)

Part	**PLA (g)**	**Time (normal / fast printer)**
Body	50	4.2 h / 1.7 h
Lid	16	1.3 h / 0.5 h
Clip	5	0.4 h / 0.2 h
Lens plate	2	0.1 h
All on one bed	**∼73**	**∼6 h / ∼2.5 h**
Test-fit piece (optional)	27	2.2 h / 0.9 h

4.6 Comparison with Expected Results

Table 4.4: Expected (design) vs. obtained results

Item	**Expected**	**Obtained**
Focus distance u	40 mm (lens marked $f = 45$ mm; Eq. (2.1) gives $v = -360$ mm, $m = 9$)	22 mm; real $f \approx 25$ mm; mirror removed
Sentence time ($n = 21$)	12.67 s scroll + 2 s gap = 14.67 s	14.6 s measured
3V3 rail	3.2–3.4 V	3.302 V
5V rail on USB	4.8–5.2 V	5.162 V
I2C address	0x3C	0x3C on both OLEDs
mDNS discovery	`glasses.local` resolves	Works; sometimes slow, IP fallback used
Clip snap strain	below PLA limit (∼2.5%)	2.2% (model check)

4.7 Challenges and Solutions

4.7.1 Technical Challenges

- The lens did not behave like its marking ($f = 45$ mm): the image was sharp at 22 mm, not 40 mm.
- The ESP32-C3 SuperMini often failed to connect at full Wi-Fi transmit power.
- The ESP32-C3 cannot join 5 GHz hotspots.
- `glasses.local` (mDNS) was sometimes not found from the laptop.

- A 4-line word-wrap made sentences cramped and hard to read on a 128×64 screen.
- The simulator first sent ISL glosses ("TODAY HOT"), so words like "is" never appeared.

4.7.2 Practical Challenges

- The stock TP4056 charges at 1 A, which is too high for a 600 mAh cell.
- Arduino's build tool (`ctags`) is x86-only and failed on the Apple Silicon laptop ("bad CPU type in executable").
- Port 8000 on the laptop was already in use by another application.
- Loose friction-fit headers gave random resets and "no I2C device" errors.
- Board and lens sizes were not all measured before the enclosure design.

Table 4.5: Problems and the actions taken

Problem	**Action taken**
Lens focus at 22 mm, not 40 mm	Mirror removed; OLED rails at 19/22/25 mm; enclosure v3
Wi-Fi fails at full TX power	`WiFi.setTxPower(WIFI_POWER_8_5dBm)`
5 GHz hotspot	2.4 GHz hotspot ("Maximize Compatibility" on iPhone)
mDNS not always found	IP shown on the OLED; `--glasses <IP>` / `--host <IP>` option
Cramped sentences	One-line, right-to-left scroll, 2 passes, then blank
Simulator showed glosses	Simulator sends the English sentence word by word
1 A charge current	USB port taped; bench supply 4.20 V / 0.30 A, stop at $\leq$ 0.05 A
`ctags` "bad CPU type"	Installed Rosetta 2
Port 8000 busy	App started with `--port 9000`
Loose headers	Press the board down first; solder only in the final build
Unmeasured part sizes	0.3 mm pocket clearance; print the test-fit piece first

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

- A working, low-cost chain from an ISL signer to text in a wearable HUD was built and tested.
- All hardware bring-up tests (Steps 1–11) passed.
- The display firmware finds the laptop by name, centres short words and scrolls sentences.
- The end-to-end demo works: camera → recognizer → Wi-Fi → glasses.
- The parametric clip-on enclosure is print-ready (∼73 g PLA).
- Measured values match the design model, e.g. sentence timing within 0.1 s.

5.2 Limitations and Assumptions

- The lens focal length ($f \approx 25\,\text{mm}$) is estimated from the focus distance, not measured.
- Glasses-side latency, end-to-end delay and recognition accuracy have not been measured yet.
- Battery runtime was tested but the value was not logged.
- The enclosure is designed and exported but not yet printed or assembled.
- The firmware has no "NO LINK" screen and no automatic Wi-Fi reconnect yet.
- The system needs a laptop and a 2.4 GHz phone hotspot; it recognises isolated words, not continuous signing.

5.3 Future Work

- Show "NO LINK" after 10 s without packets, and reconnect to Wi-Fi automatically.
- Measure the glasses-side latency (packet in → OLED update) and the end-to-end

delay.

- Measure the lens focal length directly (Step 12, O5), and check readability in daylight.
- Print, solder and assemble the final unit (Steps 13–14), and measure the battery runtime.
- Report the recognition accuracy of the model on a held-out test set.
- Add the ESP32-CAM on the glasses as the camera, so the laptop camera is not needed.

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APPENDIX A

WIRING AND BILL OF MATERIALS

A.1 Wiring Table

Table A.1: Display unit wiring

From	To
LiPo red / black	TP4056 B+ / B− (soldered)
TP4056 OUT+	Switch middle pin; switch side pin → ESP32-C3 5V
TP4056 OUT−	ESP32-C3 G
ESP32-C3 3V3	OLED VDD (VCC)
ESP32-C3 G	OLED GND
ESP32-C3 GPIO8	OLED SDA
ESP32-C3 GPIO9	OLED SCK (SCL)

A.2 Bill of Materials

- ESP32-C3 SuperMini ×2, 0.96” SSD1306 OLED ×2, TP4056 Type-C board ×2, slide switch ×3.
- LiPo 3.7 V 600 mAh (602030) ×2, 25 mm lens pair, 1 mm clear acrylic ×4.
- Breadboard, berg strips, jumper wires, soldering kit with multimeter, Type-C cable.
- About 73 g of black PLA for the enclosure.

APPENDIX B

FIRMWARE SOURCE EXCERPT

Main display functions of `firmware/glasses_display/glasses_display.ino`:

```
// New text: fits on the screen -> static in the centre; wider -> start scrolling.
void showText(const String &text) {
  int16_t x1, y1;
  uint16_t w, h;
  display.setTextSize(2);
  display.getTextBounds(text, 0, 0, &x1, &y1, &w, &h);
  if (w <= display.width()) {
    scrollText = "";
    display.clearDisplay();
    drawCentered(text, 2, (display.height() - 16) / 2);
    display.display();
  } else {
    scrollText = text;
    scrollW = w;
    scrollStart = millis();
    lastFrame = 0;
  }
}

void loop() {
  int len = udp.parsePacket();
  if (len > 0) {
    len = udp.read(packet, sizeof(packet) - 1);
    packet[len] = '\0';
    showText(jsonText(String(packet)));
  }
  if (scrollText.length() && millis() - lastFrame >= FRAME_MS) {
    lastFrame = millis();
    scrollFrame();
  }
}
```

APPENDIX C

UDP INTERFACE AND TEST TOOLS

- Glasses: mDNS name `glasses.local`, UDP port 4210 (from `config.yaml`).
- One datagram per word or sentence, UTF-8 JSON, under 256 bytes:

```
{"text": "hello", "conf": 0.92}
```

- `text`: word or short line to show; `conf`: model confidence, 0–1, 2 decimals.
- Test without the recognizer: `python tools/send_test.py --text hello`
- Simulated signing: `python software/run.py --simulate --port 9000`
- Source code: `https://github.com/Charan20510/sign_lang_hud_glasses`

APPENDIX D

BRING-UP TEST SKETCHES

- `t1_blink`: board alive (LED + Serial).
- `t2_i2c_scan`: finds the OLED at 0x3C.
- `t3_oled_hello`: big "HELLO" + counter.
- `t4_wifi_udp`: text from the laptop over Wi-Fi.
- `t5_icons`: bitmap image on the OLED.
- `t6_mirror_center`: mirrored, centred text for the HUD view.

Full step-by-step guide: `docs/hardware_bringup.md`; run guide: `docs/run_guide.md`.